

Prerak Patel

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Machine Learning Engineer | Deep Learning | Python

Melbourne, Florida | Permanent Resident, U.S. Person

SUMMARY

Machine Learning Engineer with hands-on experience building end-to-end ML workflows from data ingestion and preprocessing to training, evaluation, and scalable execution. Strong in **Python**, **PyTorch**, time-series forecasting, and generative modeling (VAE, Diffusion, Conditional Flow Matching) on large geospatial and radar/satellite datasets. Delivered reproducible pipelines with experiment tracking, multi-GPU training on Slurm HPC, and latency/performance improvements in backend systems. Comfortable working across research-to-product engineering with **AWS**, **Docker**, and modern ML tooling.

SKILLS

Languages: Python, C++, Java, SQL, JavaScript

Machine Learning: Machine Learning, Deep Learning, Supervised Learning, Time-Series Forecasting, NLP, Sentiment Analysis, Feature Engineering, Model Evaluation

Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn

Data & Visualization: NumPy, Pandas, Polars, Dask, SciPy, Statsmodels, Matplotlib, Plotly, OpenCV

MLOps / DevOps: AWS, Docker, Kubernetes, Git, CI/CD, MLflow, Hugging Face Spaces, Gradio

Backend: Django, RESTful APIs, Redis, PostgreSQL

Compute: Slurm, GPU Clusters, Multi-GPU Training

WORK EXPERIENCE

Florida Institute of Technology, Melbourne, FL

April 2026 – Present

Research Assistant

Advisor: Dr. Yang

- Building **Python** ML and time-series pipelines for cryptocurrency and stock market forecasting using NLP preprocessing, feature engineering, regression/classification, and model evaluation.
- Performing financial sentiment analysis on news/social/financial text to study relationships between sentiment signals and asset price movement.

Florida Institute of Technology, Melbourne, FL

May 2025 – Present

Graduate Research Assistant

Advisor: Dr. Georgios C. Anagnostopoulos

- Built scalable data pipelines to download, organize, and preprocess ~10 years of IMERG precipitation and IR satellite data into analysis-ready geospatial-temporal tensors, reducing manual preparation effort by 70%.
- Designed and trained **PyTorch** generative models (VAE, Diffusion, Latent Diffusion) for nowcasting/flash-flood prediction, compressing high-dimensional sequences and reducing storage/memory by 35%.
- Implemented end-to-end multimodal forecasting workflows (preprocessing, training, checkpointing, evaluation, experiment logging) and executed multi-GPU **Slurm** runs to accelerate iteration by 50%.
- Implemented FlowCast radar nowcasting benchmark using Conditional Flow Matching with multi-GPU/multi-node Slurm workflows for scalable model training and evaluation.

Matrixhive Technologies Pvt. Ltd., Ahmedabad, India

January 2024 – April 2024

Python Developer Intern

- Developed Django backend modules and RESTful APIs for client-facing workflows; delivered 5+ production features across authentication, dashboards, forms, and admin tools.
- Optimized database queries and integrated Redis caching, reducing API latency by 25% and improving application responsiveness.

PROJECTS

PKCast: Probabilistic Radar Nowcasting with Conditional Flow Matching

[GitHub Repository](#)

2026

- Developed a **PyTorch**-based probabilistic radar nowcasting system for the SEVIR VIL dataset to forecast future storm activity from historical radar sequences.
- Built a VAE-style autoencoder, generated HDF5 latent datasets, and trained a Conditional Flow Matching model with a Cuboid Transformer U-Net backbone for latent-space forecasting.
- Implemented distributed GPU training, MLflow tracking, streaming metrics, Cartopy visualizations, and **Slurm** scripts for scalable cluster training.

3D Variational Autoencoders for Satellite IMERG and IR Data

[GitHub Repository](#)

2025

- Designed and trained 3D β -VAE models on large-scale NASA IMERG precipitation and IR satellite data for spatiotemporal representation learning.
- Processed over 1 TB of geospatial data and optimized multi-GPU HPC workflows, reducing memory usage by 35% and accelerating training by 50%.

Satellite Image Segmentation for Environmental Monitoring

[GitHub Repository](#) | [Live Demo](#)

2024

- Built and deployed a U-Net-based satellite image segmentation system; achieved 86% segmentation accuracy and deployed an interactive Gradio web interface.

EDUCATION

Master of Science in Computer Engineering Florida Institute of Technology Melbourne, FL	January 2025 – December 2026
Bachelor of Engineering in Computer Engineering Gandhinagar Institute of Technology Gandhinagar, Gujarat, India	May 2020 – May 2024